

DGX销售培训



Stephen Wu

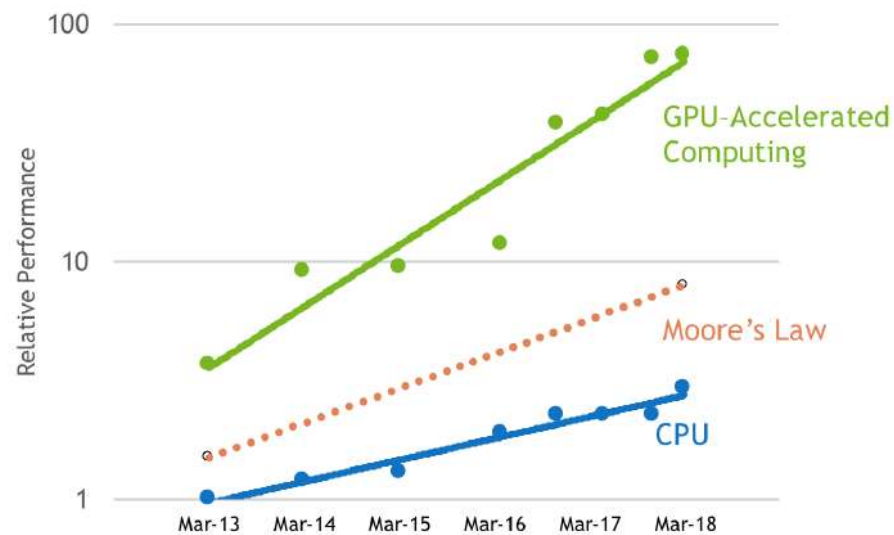
超越摩尔定律

2013

cuBLAS: 5.0
cuFFT: 5.0
cuRAND: 5.0
cuSPARSE: 5.0
NPP: 5.0
Thrust: 1.5.3
CUDA: 5.0
Resource Mgr: r304
Base OS: CentOS 6.2



Accelerated Server
With Fermi



Measured performance of Amber, CHROMA, GTC, LAMMPS, MILC, NAMD, Quantum Espresso, SPECfem3D

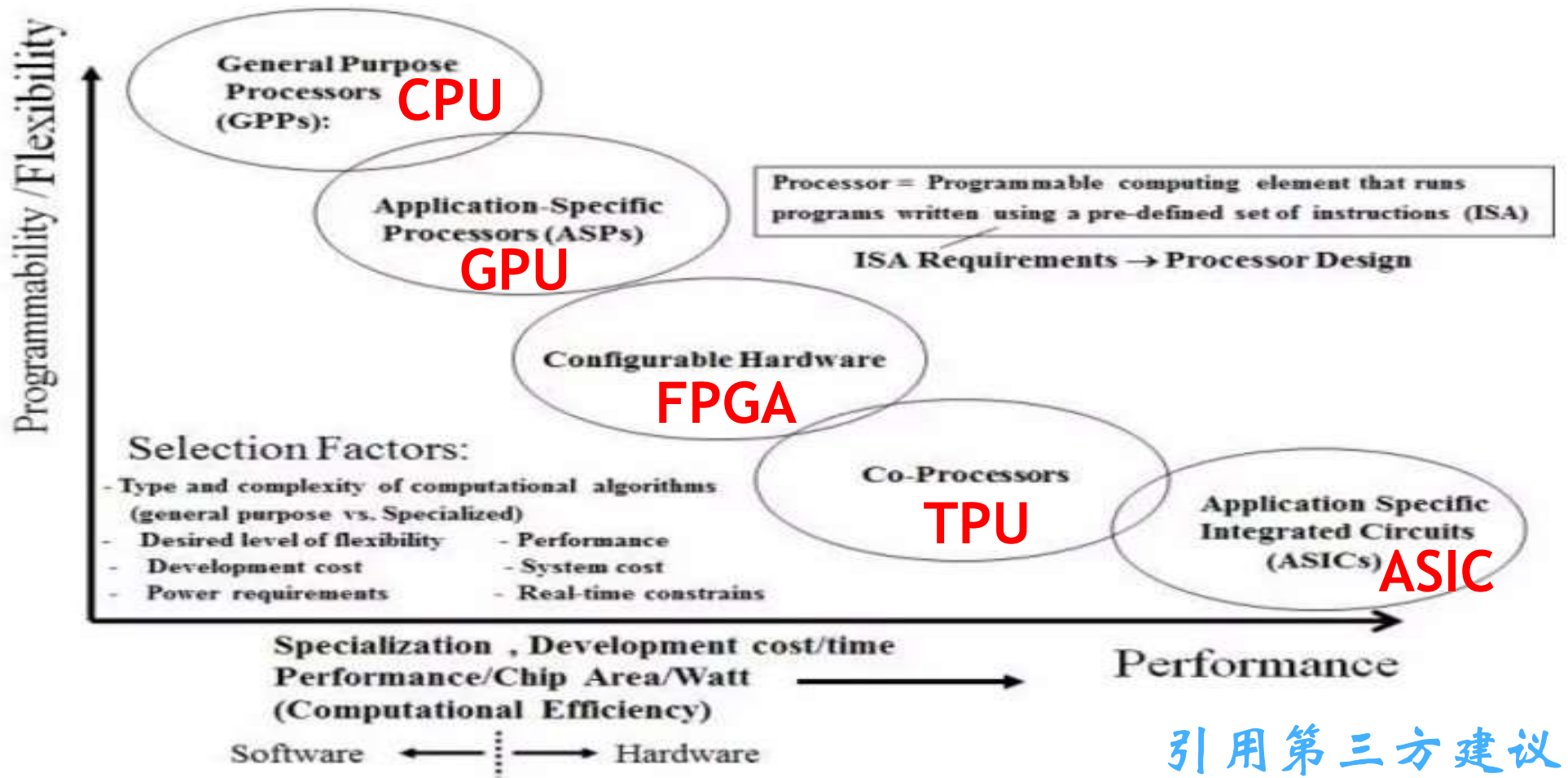
2018

cuBLAS: 9.0
cuFFT: 9.0
cuRAND: 9.0
cuSOLVER: 9.0
cuSPARSE: 9.0
NPP: 9.0
Thrust: 1.9.0
CUDA: 9.0
Resource Mgr: r384
Base OS: Ubuntu 16.04

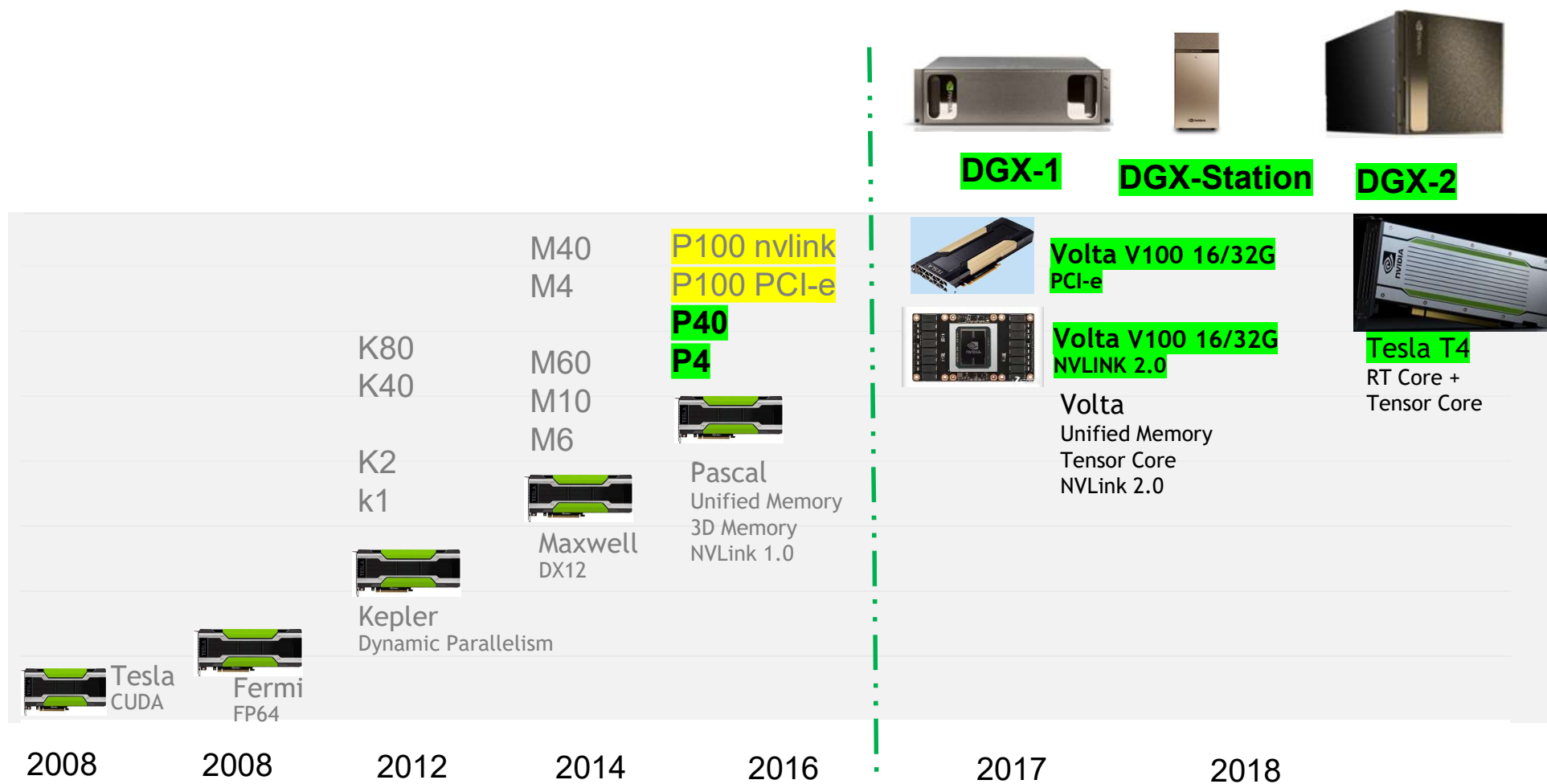


Accelerated Server
with Volta

1 不同类型AI服务器之比较



TESLA 路线图



TESLA V100 NV-LINK

5,120 CUDA cores

640 NEW Tensor cores

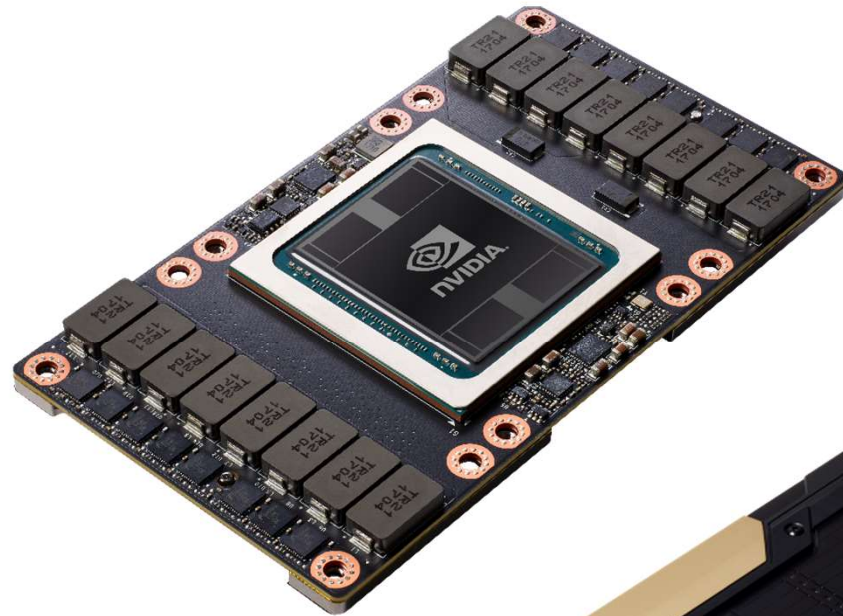
7.8 FP64 TFLOPS | 15.7 FP32 TFLOPS

125 Tensor TFLOPS

20MB SM RF | 16MB Cache | 32GB

HBM2 @ 900 GB/s

300 GB/s NVLink



Tesla V100 PCIe

5,120 CUDA cores | 640 Tensor cores

7.0 FP64 TFLOPS | 14 FP32 TFLOPS

NEW 112 Tensor TFLOPS

20MB SM RF | 16MB Cache | 32GB

16GB HBM2 @ 900 GB/s

32 GB/s PCIe Gen3



TESLA T4

WORLD'S MOST ADVANCED INFERENCE GPU

Universal Inference Acceleration

320 Turing Tensor cores

2,560 CUDA cores

8.1 FP32 TFLOPS

65 FP16 TFLOPS | 130 INT8 TOPS | 260 INT4 TOPS

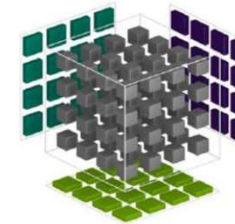
16GB | 320GB/s



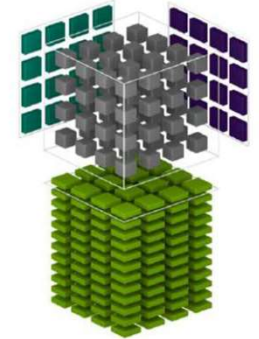
TENSOR CORE

Mixed Precision Matrix Math
4x4 matrices

Pascal



Volta Tensor 核心



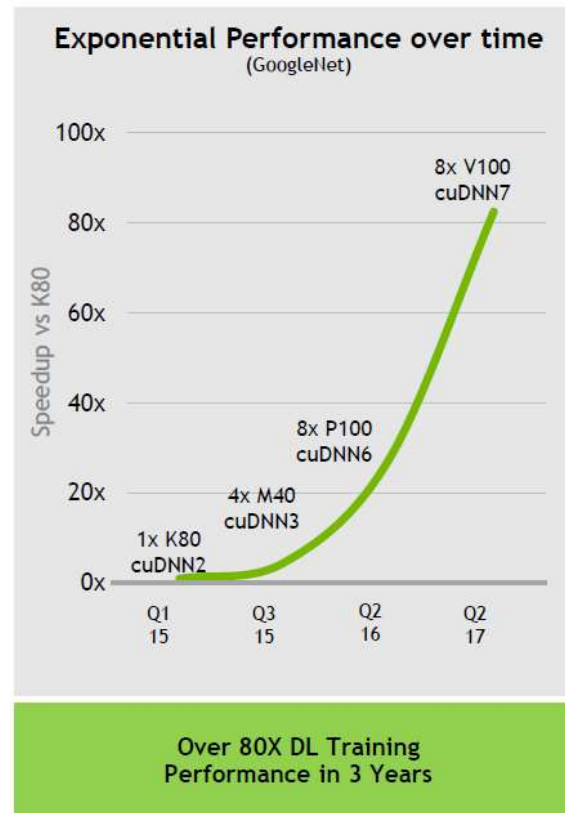
$$\mathbf{D} = \begin{pmatrix} A_{0,0} & A_{0,1} & A_{0,2} & A_{0,3} \\ A_{1,0} & A_{1,1} & A_{1,2} & A_{1,3} \\ A_{2,0} & A_{2,1} & A_{2,2} & A_{2,3} \\ A_{3,0} & A_{3,1} & A_{3,2} & A_{3,3} \end{pmatrix} \begin{pmatrix} B_{0,0} & B_{0,1} & B_{0,2} & B_{0,3} \\ B_{1,0} & B_{1,1} & B_{1,2} & B_{1,3} \\ B_{2,0} & B_{2,1} & B_{2,2} & B_{2,3} \\ B_{3,0} & B_{3,1} & B_{3,2} & B_{3,3} \end{pmatrix} + \begin{pmatrix} C_{0,0} & C_{0,1} & C_{0,2} & C_{0,3} \\ C_{1,0} & C_{1,1} & C_{1,2} & C_{1,3} \\ C_{2,0} & C_{2,1} & C_{2,2} & C_{2,3} \\ C_{3,0} & C_{3,1} & C_{3,2} & C_{3,3} \end{pmatrix}$$

FP16 or FP32 FP16 FP16 or FP32

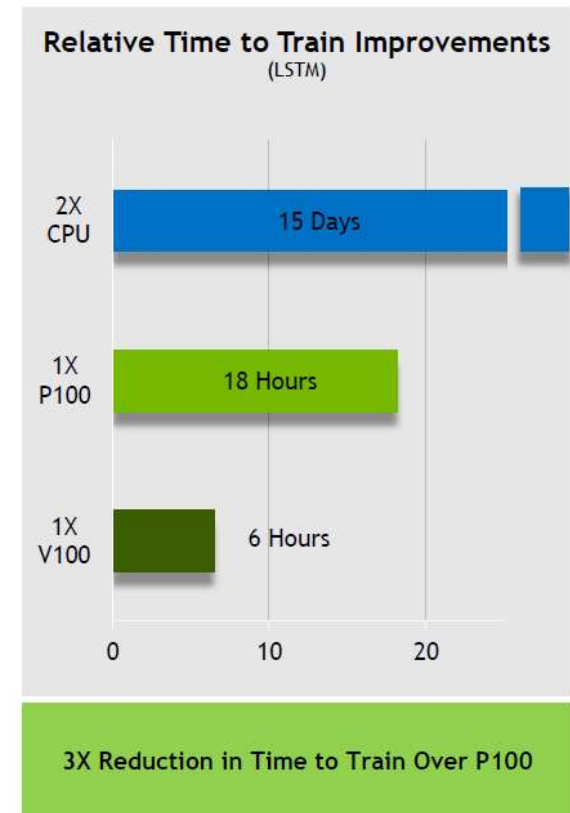
$$\mathbf{D} = \mathbf{AB} + \mathbf{C}$$

V100带来变革性的AI性能提升

3X Faster DL Training Performance



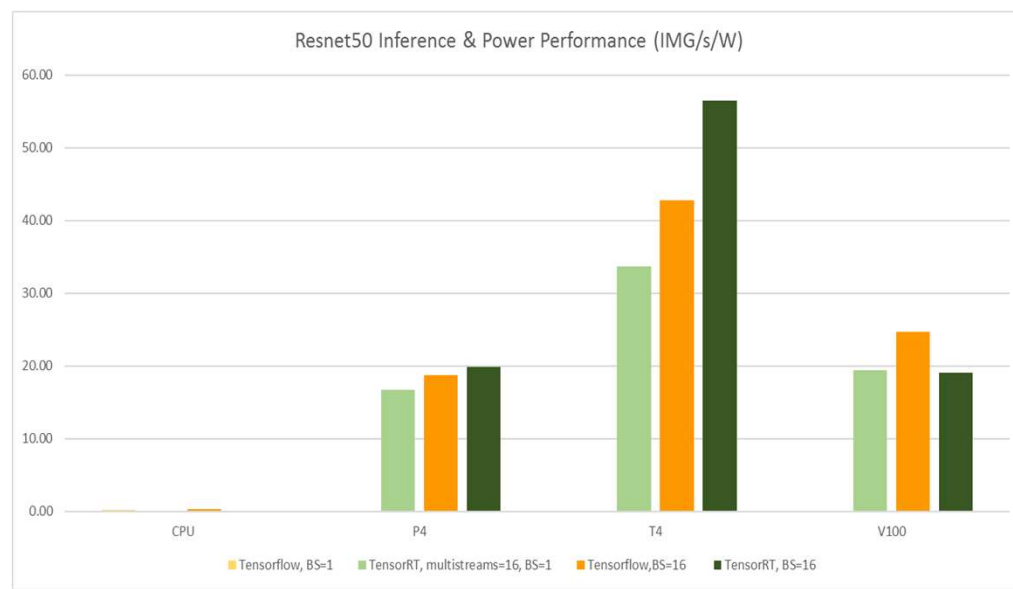
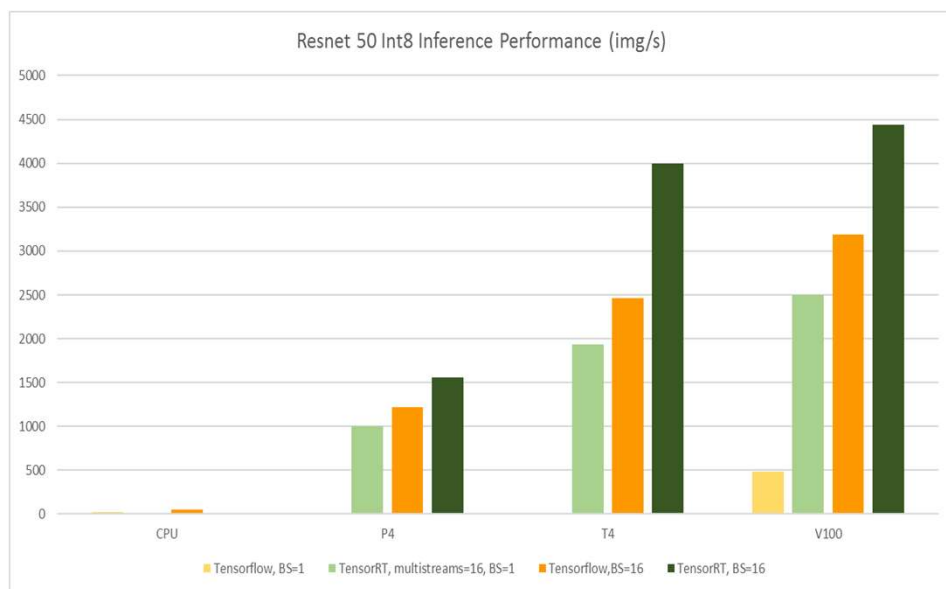
GoogleNet Training Performance on versions of cuDNN
Vs 1x K80 cuDNN2



Neural Machine Translation Training for 13 Epochs | German -> English, WMT15 subset | CPU = 2x Xeon E5 2699 V4

>2.5X INFERENCE PERFORMANCE OF P4

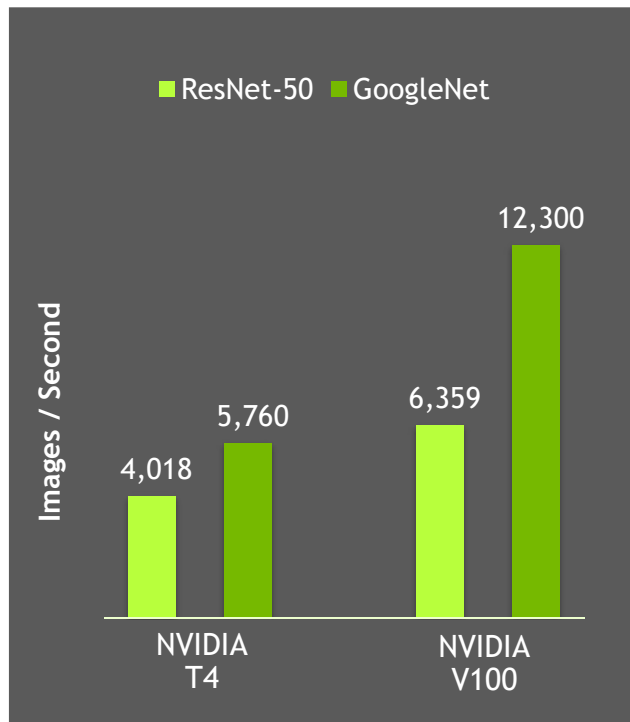
Measured across frameworks and batch sizes



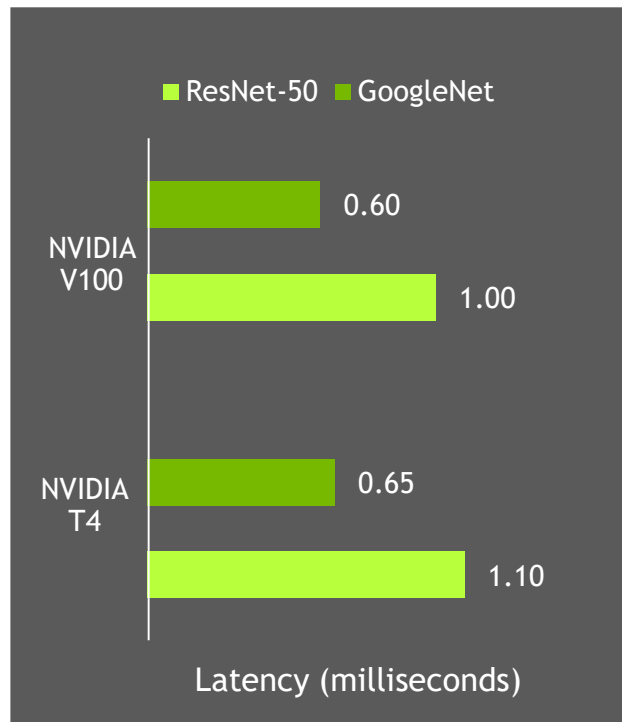
WORLD'S FASTEST INFERENCE PERFORMANCE

NVIDIA GPUs Set New Performance Records

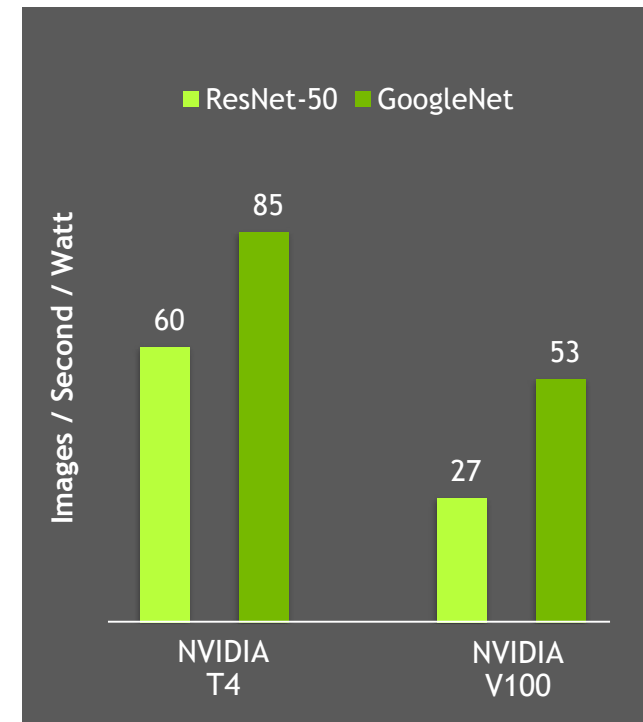
Throughput



Latency



Energy Efficiency



3 AI超级计算机DGX系统详解

NVIDIA DGX-1

世界第一台深度学习超级计算机

计算能力约为400台CPU服务器



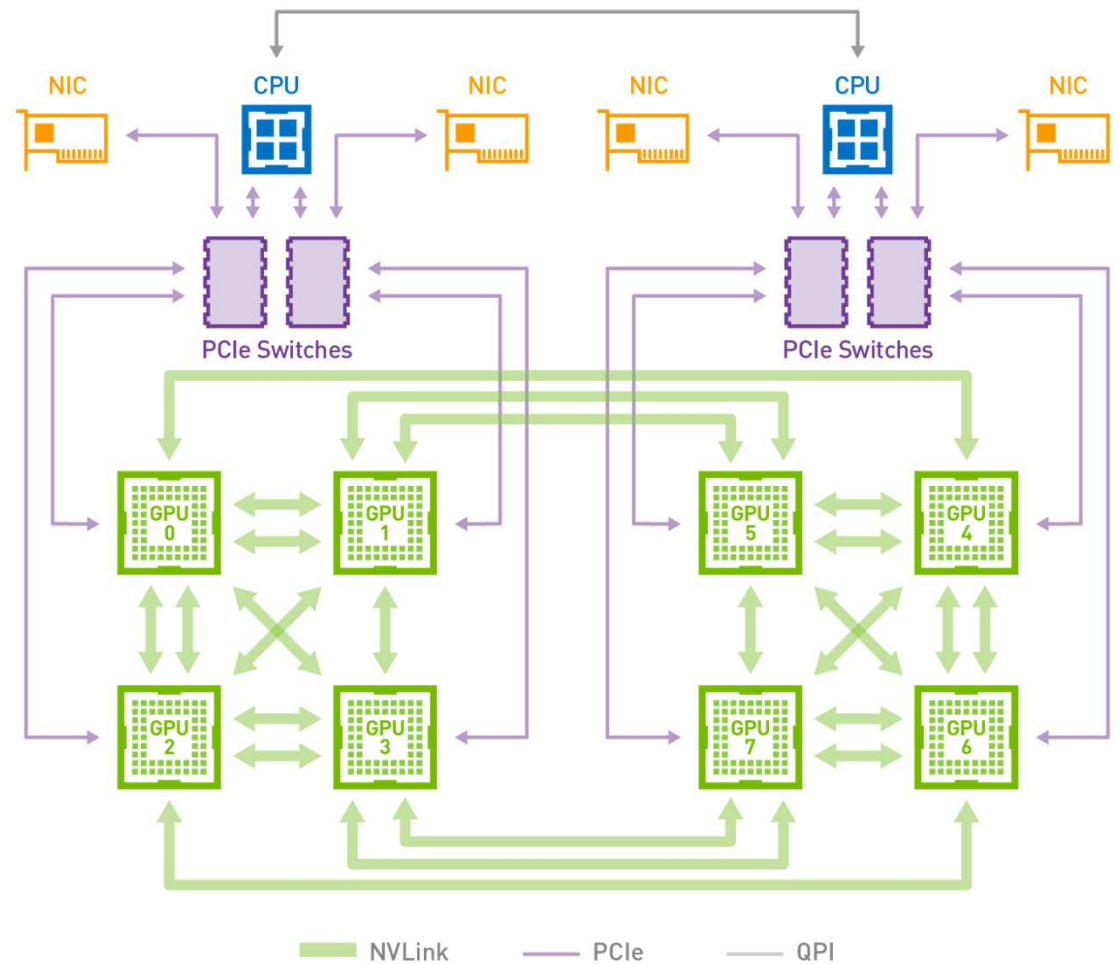
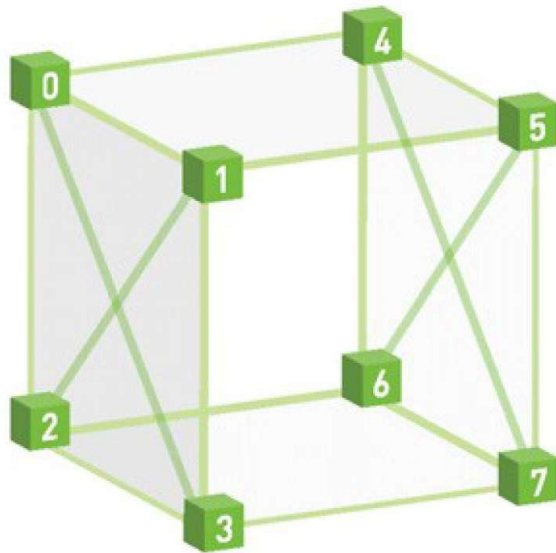
- 1 PFLOPS 深度学习计算能力
- 8张Tesla V100 GPU
- NVLink 300GB/s GPU高速互联
- 天然支持所有的深度学习框架
- 双至强CPU
- 7 TB固态硬盘
- 双10Gb以太网
- 4个100Gb Infiniband网络
- 3RU – 3200W

VOLTA NVLINK

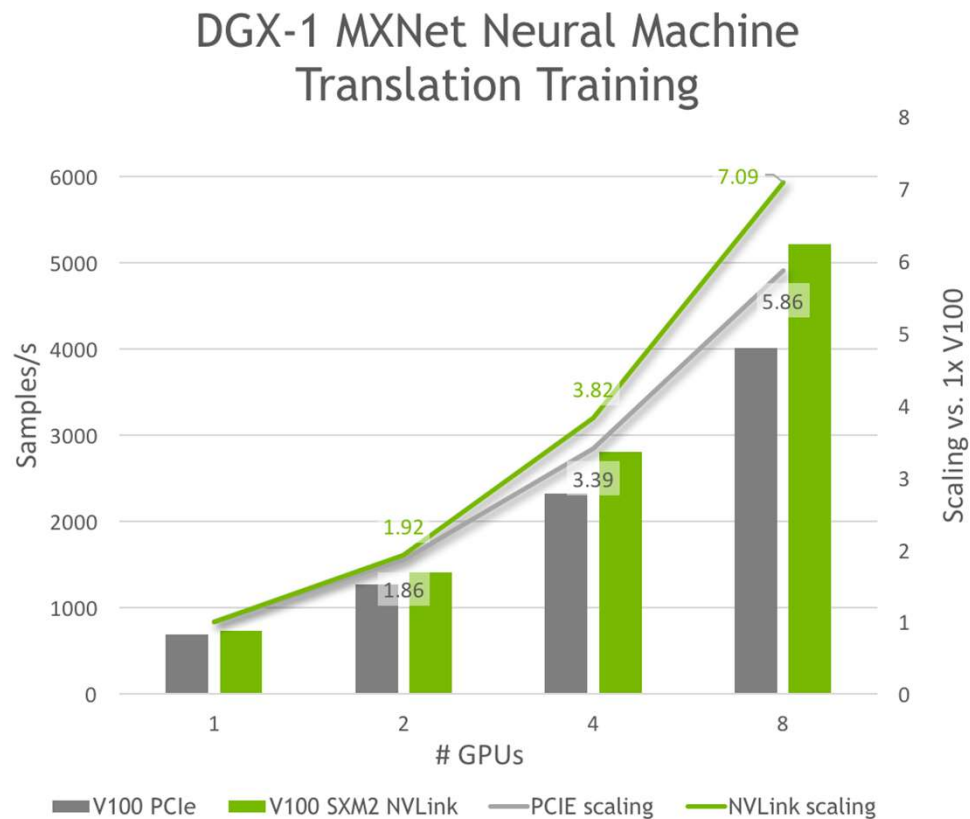
300GB/sec

50% more links

28% faster signaling



30% BETTER PERFORMANCE WITH NVLINK THAN PCIE



- Encoder and decoder embedding size of 512
- Batch size of 256 per GPU
- NVIDIA DGX containers version 17.11, processing real data with cuDNN 7.0.4, NCCL 2.1.2

世界第一的个人 AI 超级计算机 NVIDIA DGX STATION

超强计算能力，液冷无声

480 Tensor TFLOPS | 4x Tesla V100 16GB

NVLink Fully Connected | 3x DisplayPort

1500W | Water Cooled





DGX STATION SPECIFICATIONS

At a Glance

GPUs	4x NVIDIA® Tesla® V100
TFLOPS (GPU FP16)	500
GPU Memory	32 GB per GPU
NVIDIA Tensor Cores	2,560 (total)
NVIDIA CUDA Cores	20,480 (total)
CPU	Intel Xeon E5-2698 v4 2.2 GHz (20-core)
System Memory	256 GB RDIMM DDR4
Storage	Data: 3 x 1.92 TB SSD RAID 0 OS: 1 x 1.92 TB SSD
Network	Dual 10GBASE-T LAN (RJ45)
Display	3x DisplayPort, 4K Resolution
Additional Ports	2x eSATA, 2x USB 3.1, 4x USB 3.0
Acoustics	< 35 dB
Maximum Power Requirements	1500 W
Operating Temperature Range	10 - 30 °C
Software	Ubuntu Desktop Linux OS DGX Recommended GPU Driver CUDA Toolkit

NVIDIA DGX-2世界最强的超级计算机

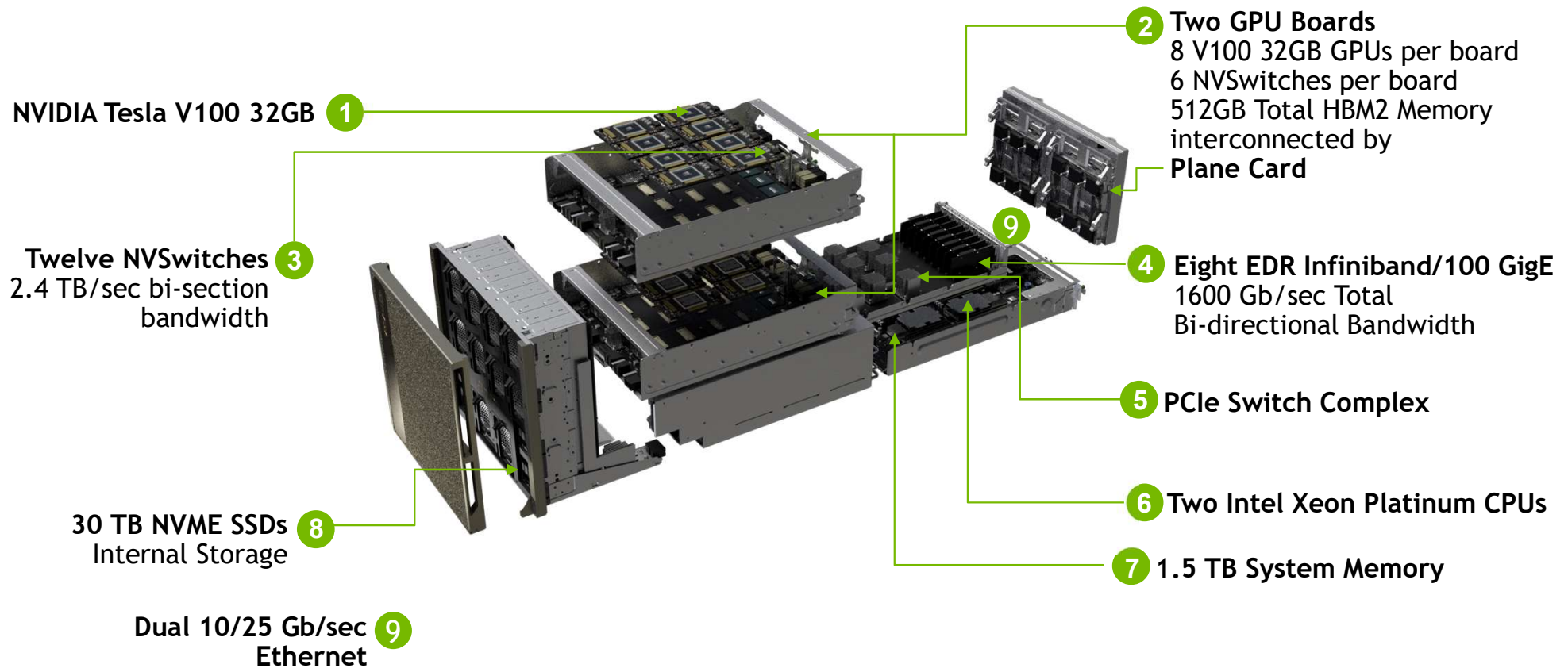
THE WORLD'S MOST POWERFUL DEEP LEARNING SYSTEM
FOR THE MOST COMPLEX DEEP LEARNING CHALLENGES

- First 2 PFLOPS System
- 16 V100 32GB GPUs Fully Interconnected
- NVSwitch: 2.4 TB/s bisection bandwidth
- 24X GPU-GPU Bandwidth
- 0.5 TB of Unified GPU Memory
- 10X Deep Learning Performance

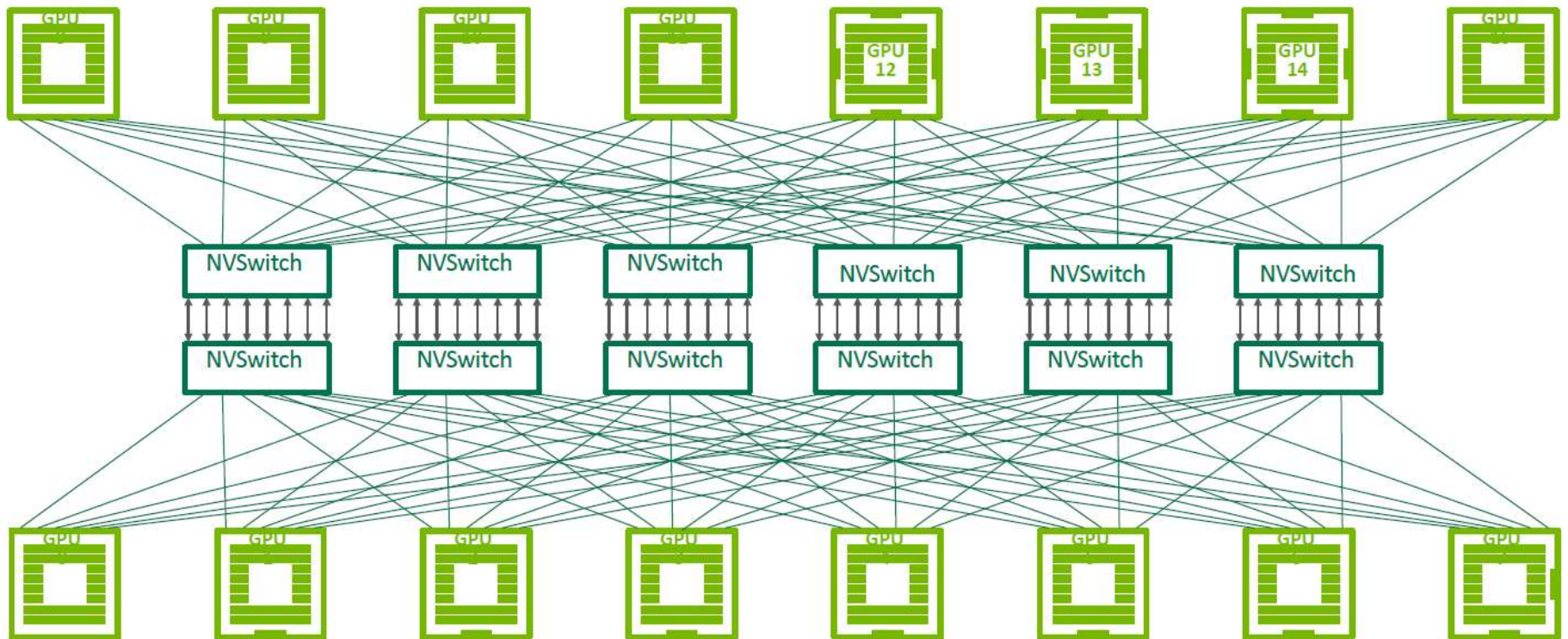


Support in CUDA 10.0

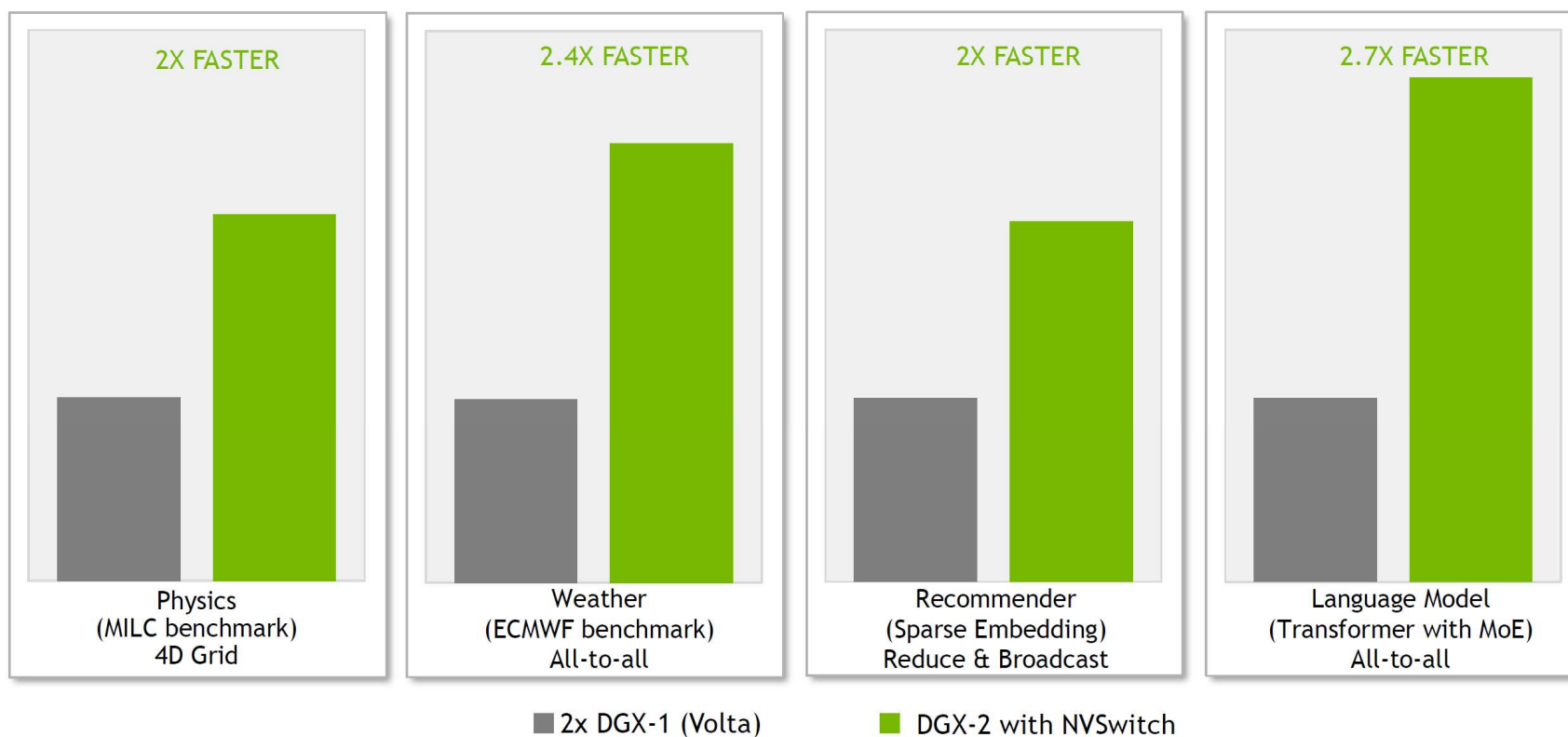
DESIGNED TO TRAIN THE PREVIOUSLY IMPOSSIBLE



FULL NON-BLOCKING BANDWIDTH



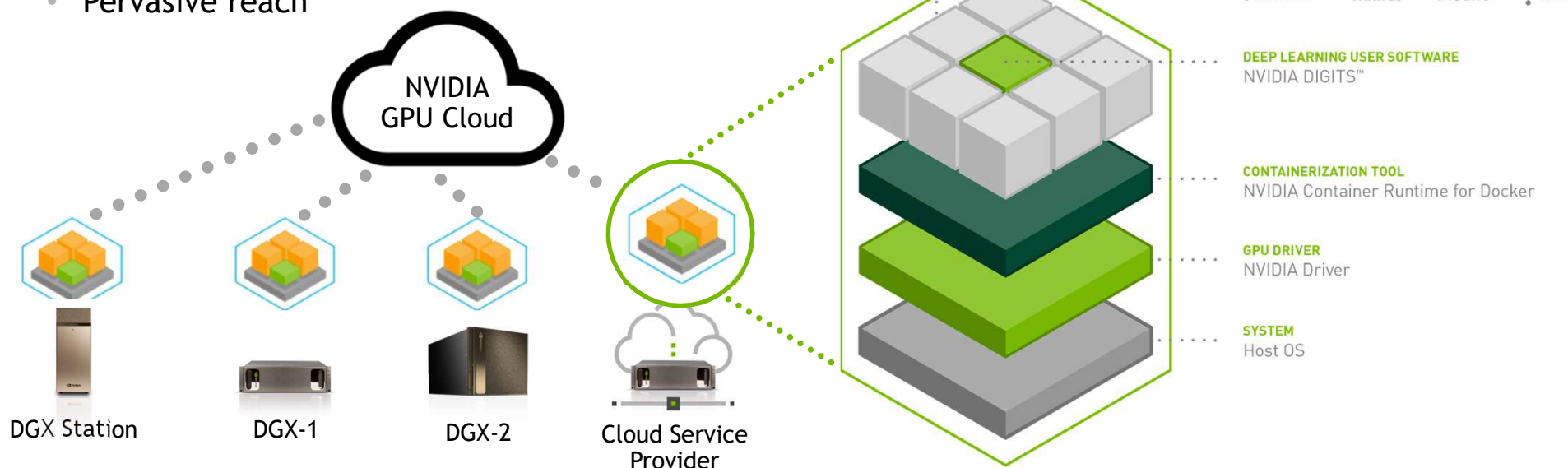
2X HIGHER PERFORMANCE WITH NVSWITCH



2 DGX-1V servers have dual socket Xeon E5 2698v4 Processor, 8 x V100 GPUs. Servers connected via 4X 100Gb IB ports | DGX-2 server has dual-socket Xeon Platinum 8168 Processor, 16 V100 GPUs

COMMON SOFTWARE STACK ACROSS DGX FAMILY

- Single, unified stack for deep learning frameworks
- Predictable execution across platforms
- Pervasive reach



NGC CONTAINER REGISTRY

10 Containers at Launch, 35 Containers Today

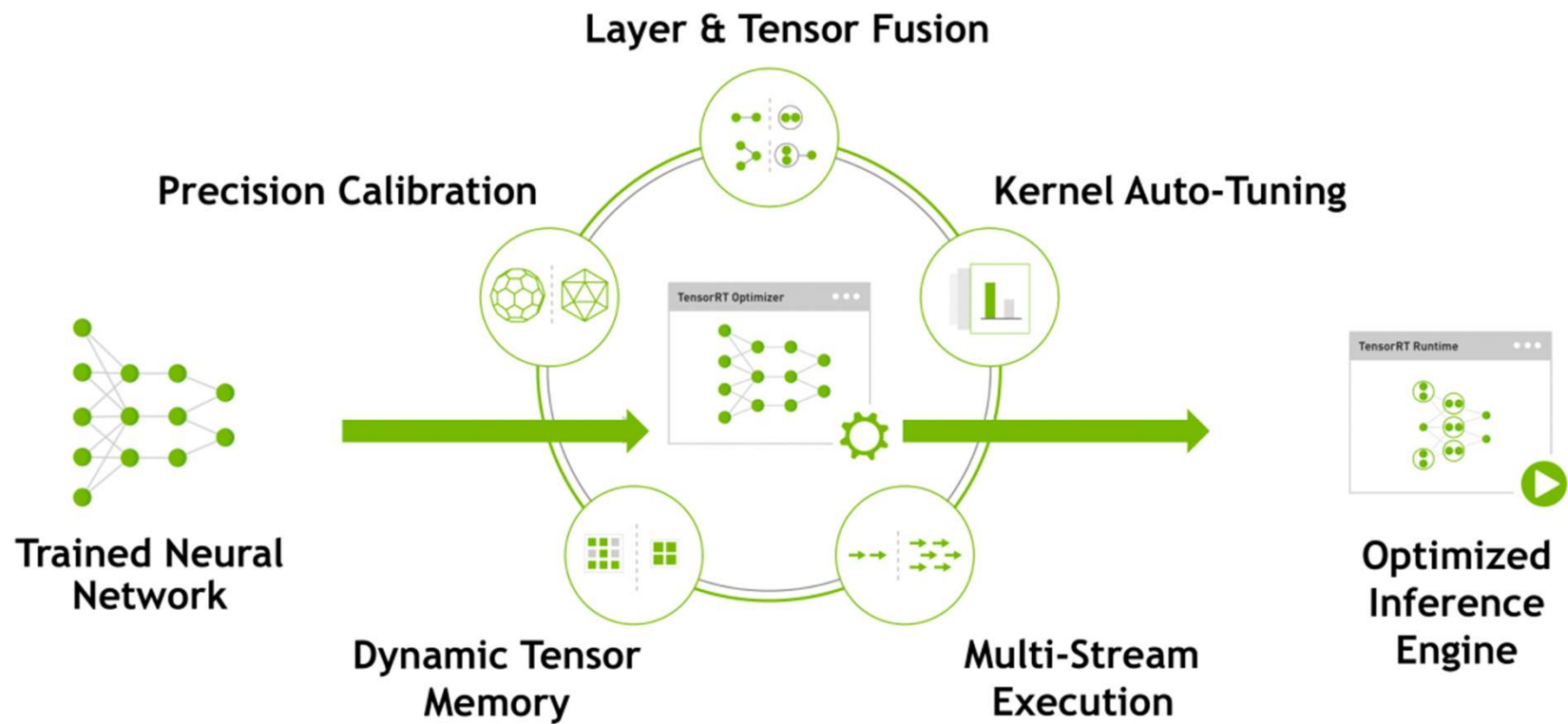
Deep Learning	HPC	HPC Visualization	NVIDIA/K8s	Partners
caffe	bigdft	index	Kubernetes on NVIDIA GPUs	chainer
caffe2	candle	paraview-holodeck		h20ai-driverless
cntk	chroma	paraview-index		kinetica
cuda	gamess	paraview-optix		mapd
digits	gromacs			paddlepaddle
inferenceserver	lammps			
mxnet	lattice-microbes			
pytorch	milc			
tensorflow	namd			
tensorrt	pgi			
theano	picongpu			
torch	relion			
	vmd			



NVIDIA
GPU CLOUD

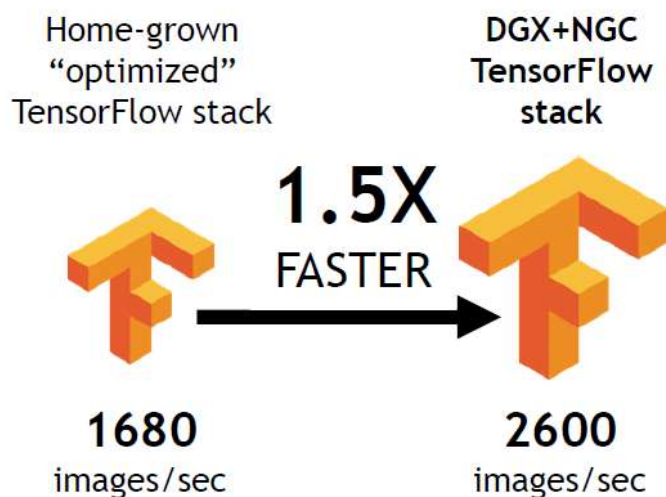
ngc.nvidia.com

TENSORRT OPTIMIZER



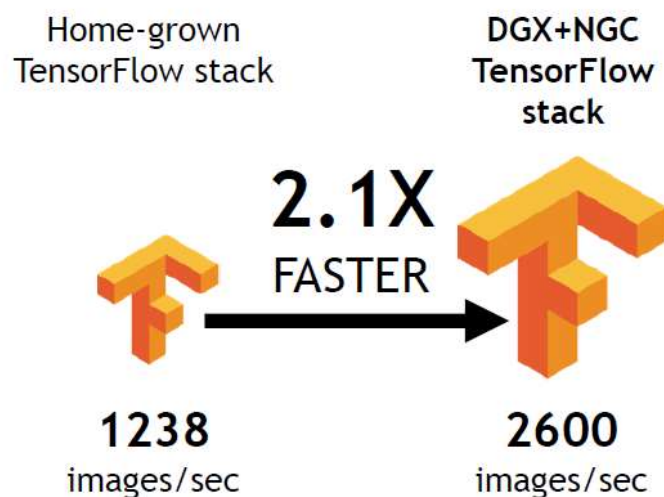
实际的性能提升

GLOBAL TECHNOLOGY FIRM
SPECIALIZING IN DIGITAL MEDIA



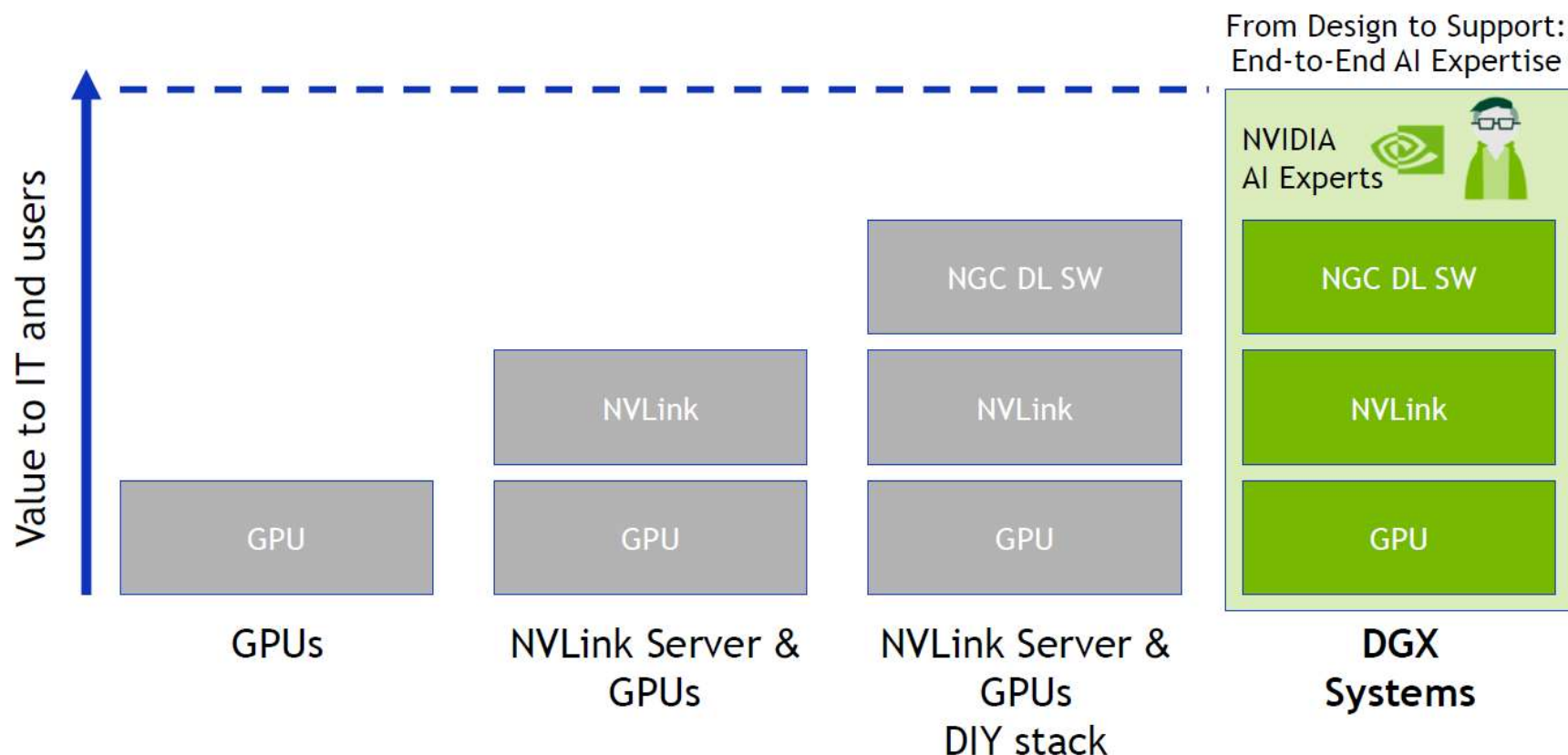
ResNet50 Training

WORLD-LEADING MEDICAL
RESEARCH CENTER



ResNet50 Training

DGX SYSTEMS – THE VALUE OF INTEGRATED SOLUTIONS AND AI EXPERTISE



AI TRAINING REQUIRES FULL STACK INNOVATION



2015



2016



2017

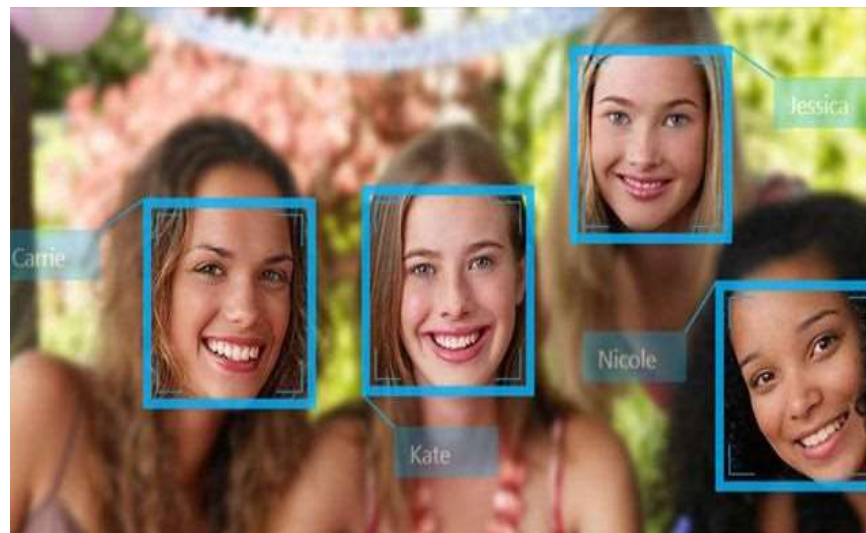


2018

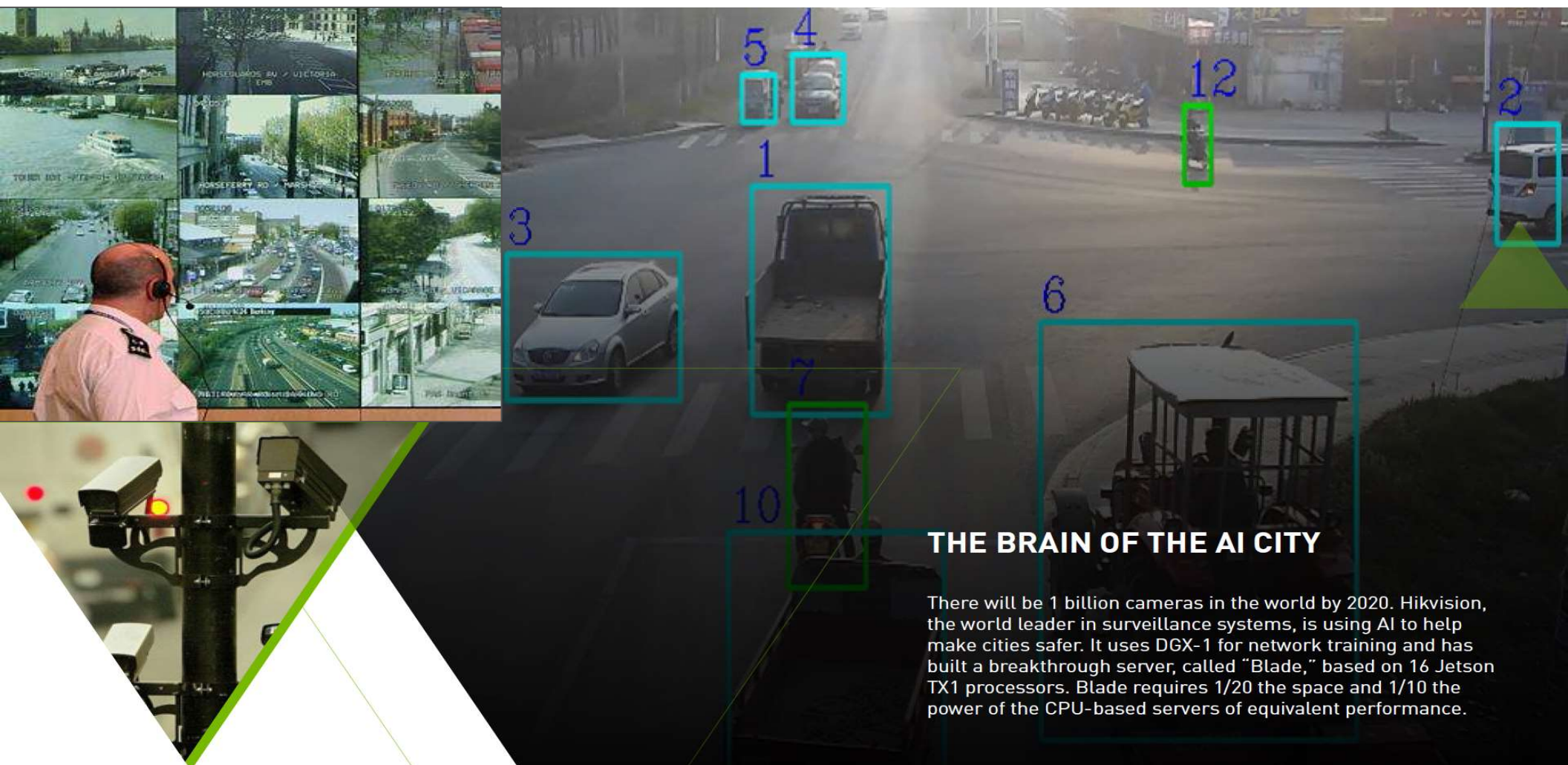


DGX系统各行业应用案例

互联网行业应用



公安行业应用



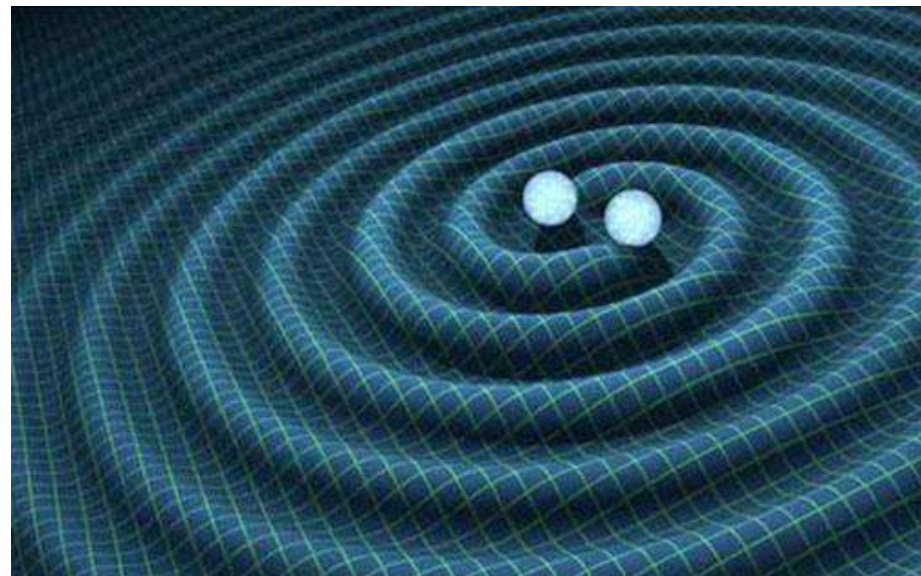
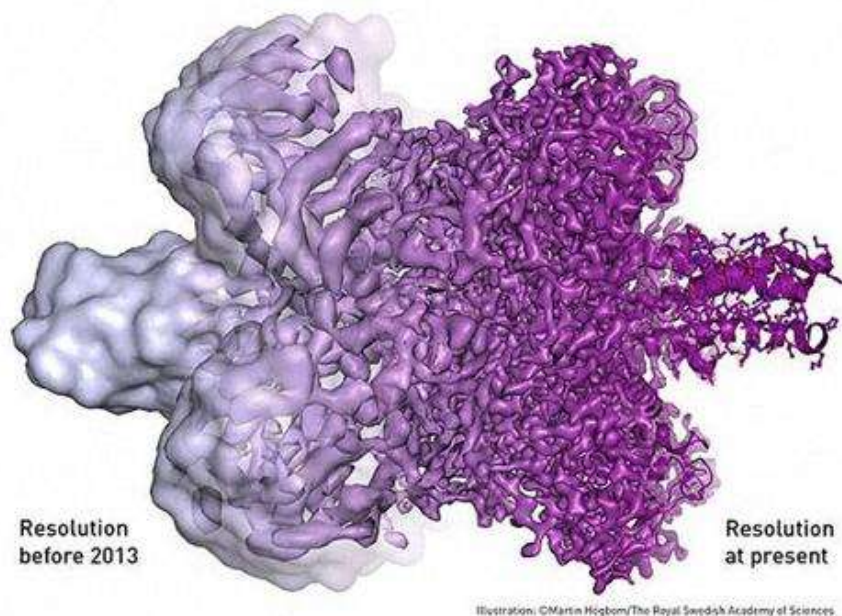
初创公司的应用



Megvii 旷视科技

教育科研行业应用

对于一个包含数百万个颗粒，即使在高性能计算集群上，也可能要花费超过50万 CPU 小时的时间。引入 GPU 加速技术是目前很多主流软件的选择，譬如Relion GPU版，在GPU的加速帮助下，已经大大缩短了分析计算的时间与成本。



近几年来，清华大学研究团队主要与麻省理工学院、加州理工学院和西澳大利亚大学等LSC成员合作开展工作，主要研究成果包括：GPU加速引力波数据分析和实现低延迟实时致密双星并合信号的搜寻；采用机器学习方法加强引力波数据噪声的分析；



制造行业应用

TCL-China Star Optoelectronics Technology |
IBM Visual Insights

- ▶ 150多名检查工作人员
- ▶ 平均检查时间20-30分钟
- ▶ DL (Visual Insights) <4分钟
- ▶ 与人类检查员一样准确

医疗影像行业的应用

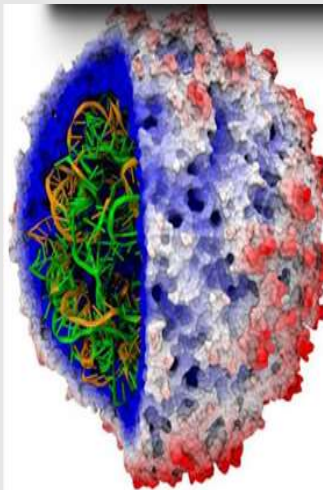
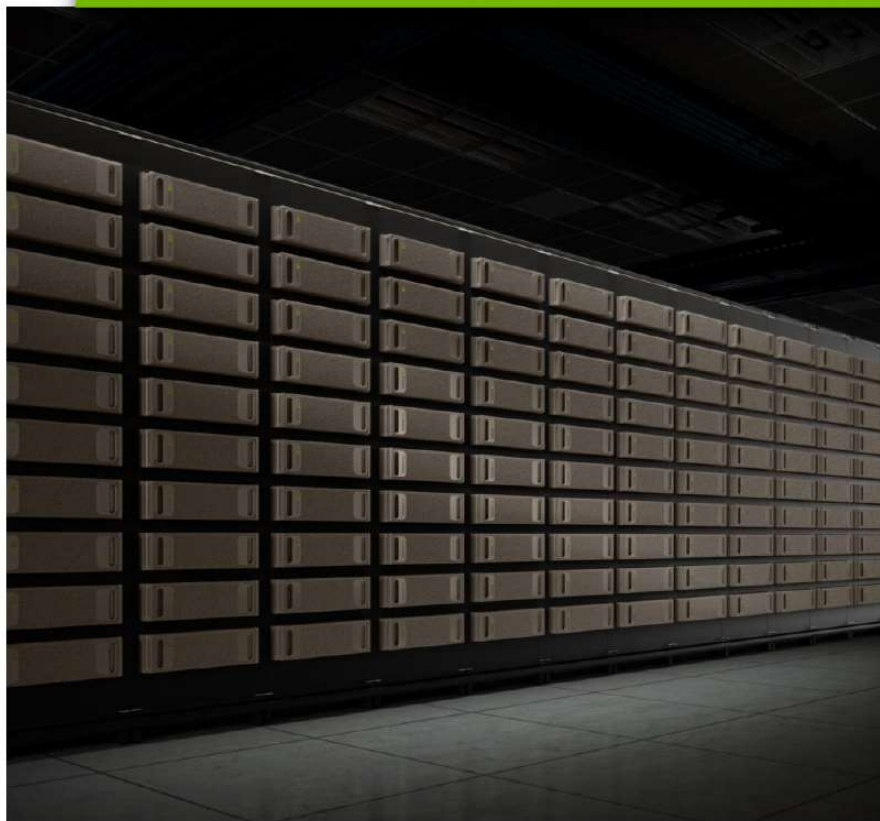
input image				
	aorta_thoracic / tortuous / mild aorta_thoracic / tortuous	opacity / lung / middle_lobe / right / aorta_thoracic / tortuous opacity / lung / base / left	calcified_granuloma / lung / middle_lobe / right / multiple calcified_granuloma / lung / hilum / right	opacity / lung / middle_lobe / right / blood_vessels calcified_granuloma / lung / middle_lobe / right
generated annotation				
true annotation				
	airspace_disease / lung / hilum / right / lung / hilum nodule / lung / hilum / right	thoracic_vertebrae_degenerative / mild aorta_tortuous / thoracic_vertebrae_degenerative / mild	normal normal	normal normal

In what may be a first for radiology images, NIH researchers trained a recurrent neural network to describe the context of diseases using chest X-rays.

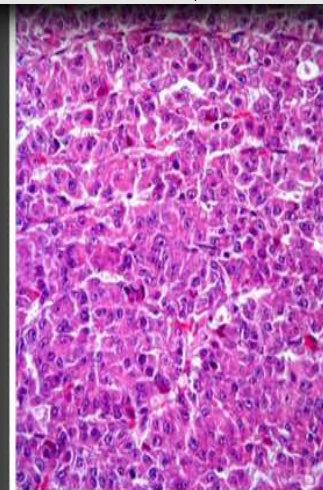
INTRODUCING DGX SATURNV

124 NVIDIA DGX-1 “Rocket for Cancer Moonshot”

癌症研究的应用



Accelerate Discovery
of Cancer Therapies



Predict Drug Response
of Cancer Patients



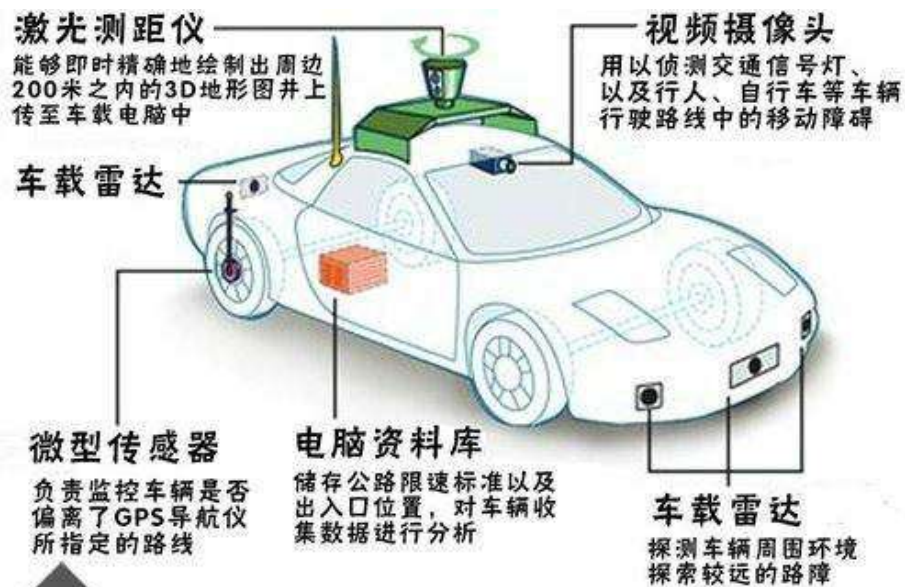
Automate Analysis
of Treatment Effectiveness

金融行业典型应用

资本市场	零售银行业务	保险
- 交易订单执行（优化交易算法，高频交易等） - 资产类组合优化 - 信贷风险（抵押风险，自动负债风险）优于信用分数 - 投资组合优化 - 你应该投资什么 - 智能顾问 - 你应该怎么用钱？	- 风险控制及反欺诈 - 信用卡风险分析 - 宏观指标分析与预估 - 智能客服中心 - 提升客户满意/保留客户 - 客户细分，智能营销	- 自动理赔的执行 - 揭露保险欺诈 - 风险量化（如医疗健康保险的中的风险控制）
跨行业： 网络安全；法规遵从；大量内存加速的SaaS数据库/数		

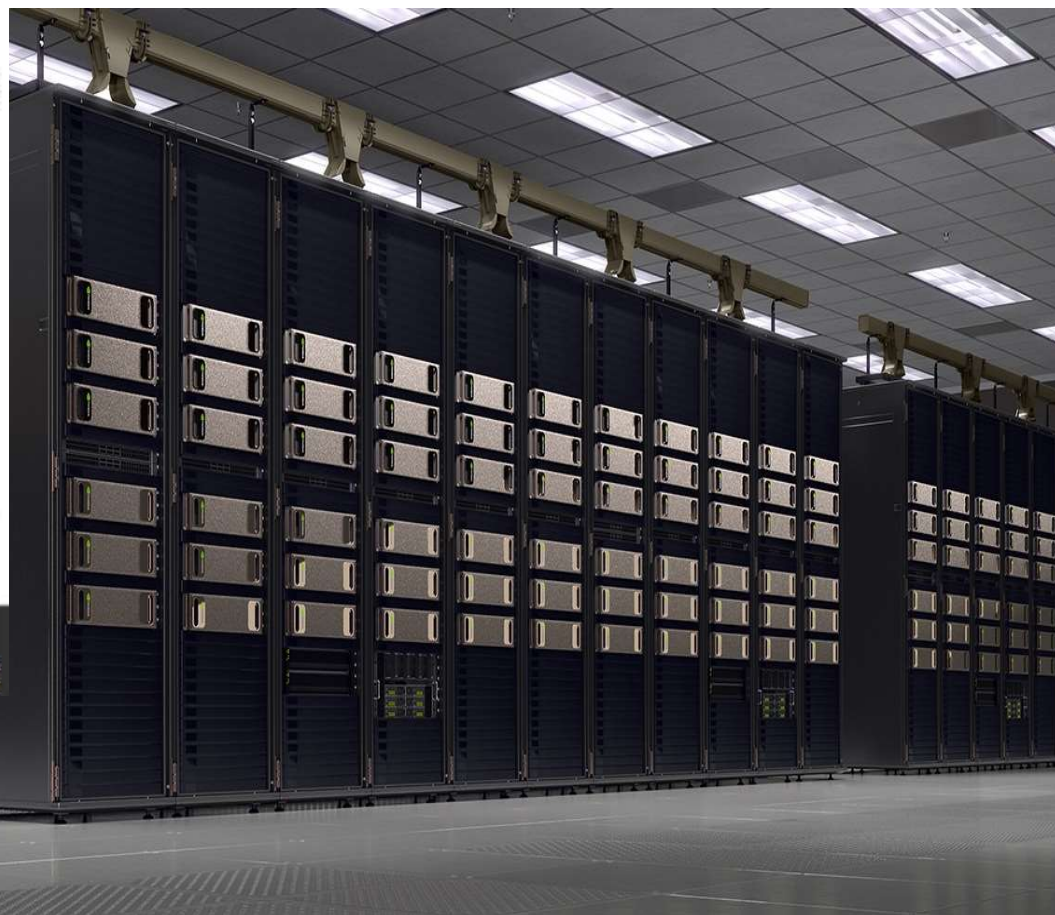


汽车行业应用



无人驾驶系统中的智能感应设备

车辆拥有全方位的智能感应设备以及系统后, 才有望达到无人驾驶的程度。 汽车之家

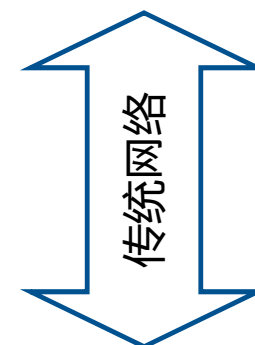


电信行业应用：移动边缘计算

Cloud
云端



MEC
雾端

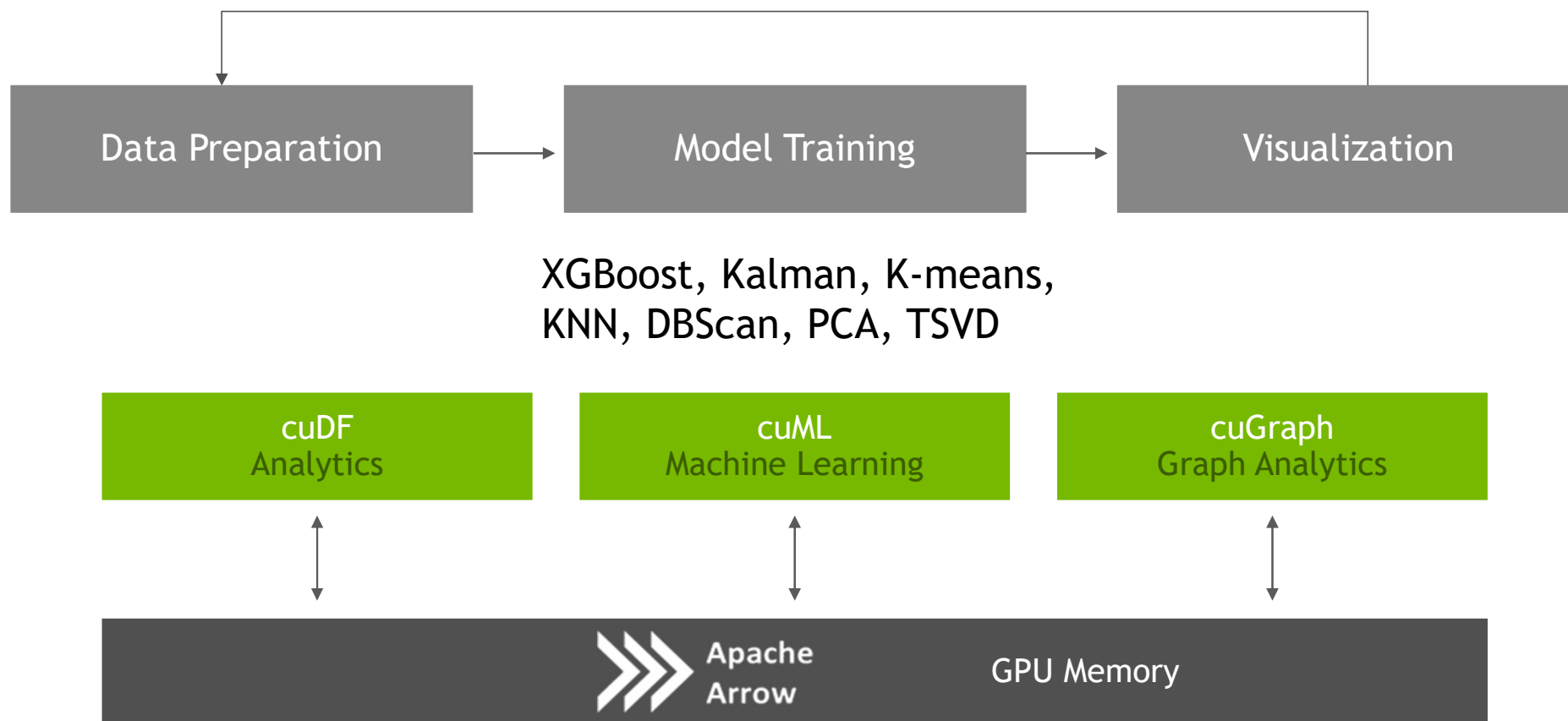


雾
智能
终端

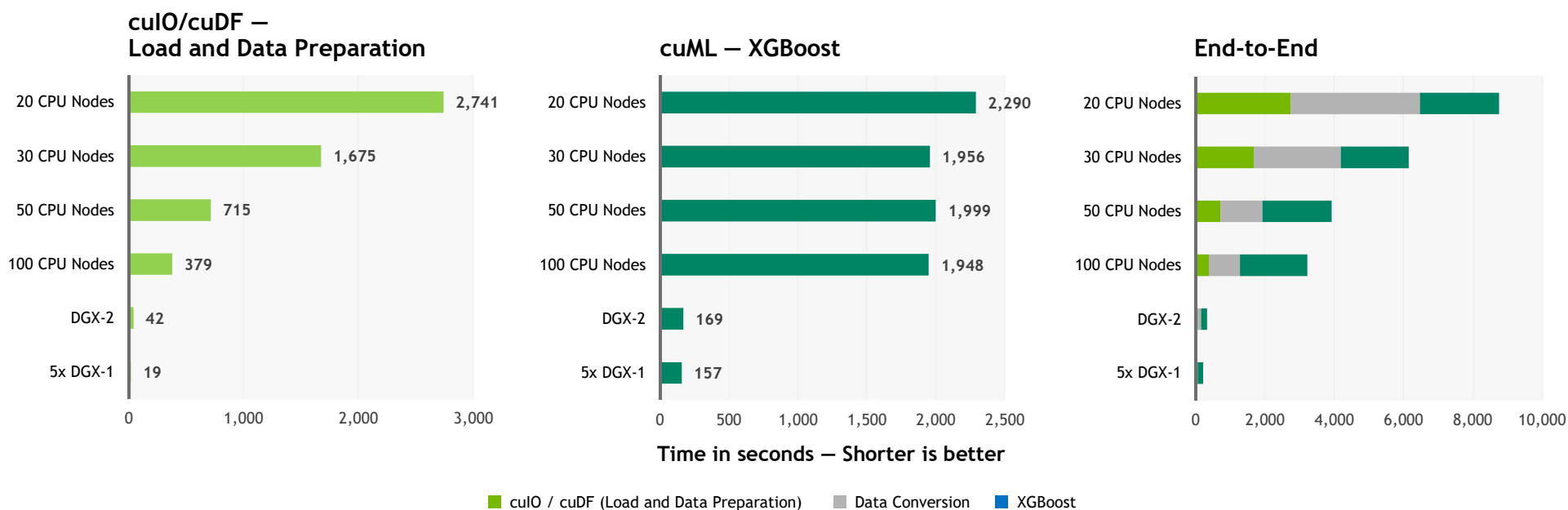


硬
智能
终端

RAPIDS 开源系统加速机器学习



FASTER SPEEDS, REAL WORLD BENEFITS



Benchmark

200GB CSV dataset; Data preparation includes joins, variable transformations.

CPU Cluster Configuration

CPU nodes (61 GiB of memory, 8 vCPUs, 64-bit platform), Apache Spark

DGX Cluster Configuration

5x DGX-1 on InfiniBand network



several hundred thousand a few more
gtcys and we'll be ready to go okay

